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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

MARKING SCHEME

NOVEMBER 2015

PHYSICS **9188/3**

1 (a) (i) kelvin (K) (Reject wrong physics) B1

(ii) $[\delta] = \dfrac{J^4K^{-4}}{(J^3s^3)(m^2s^{-2})}$ Reje W.P. C1

$= Js^{-1}K^{-4}m^{-2}$ C1

Base units of δ = kg $K^{-4}s^{-3}$ A1 Award answer mark if W.P is followed by correct answer ~~A1~~

(iii) It fails to check the presence of the constant a. Reject any other response not dealing with constant (a) B1

(b) (i) accuracy - degree by which the mean value is away from the true value AW B1

Accept idea of measure/closeness of ~~mean~~ to T.V. value

precision - degree of scatter of readings about their mean value AW B1

Accept definition with reference to instrument

(ii) random error Accept ~~de~~ idea of closeness/ spread/Scatter, B1

cause readings to be scattered about their mean. B1

(iii) Results were not accurate since their mean value of 8,63 Nkg^{-1} A1

is far from the true value of 9.81 N kg^{-1} and ~~not~~ precise. A1

Scatter of measured values is small

(c) (i) For no skidding ~~$\tan\theta = \mu$~~ (coefficient of friction). friction = centripetal force

$\mu mg = \dfrac{mv^2}{r}$

$\cancel{\tan\theta} = \dfrac{v^2}{rg} = \dfrac{15^2}{(200)(9.81)}$ C1

Accept $\mu = \dfrac{25}{218}$

$= 0.1147$

$\therefore \mu_{min} = 0.1147$ Accept 1 s.f. A1

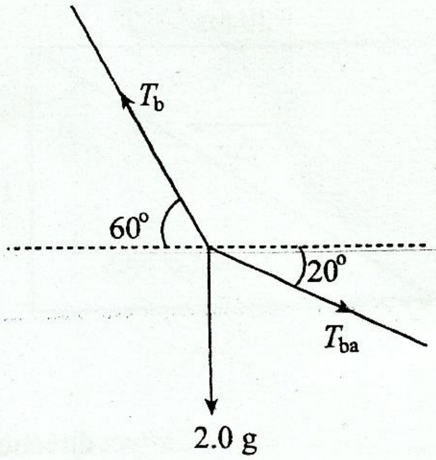
(ii) Wheel is rotating with forward velocity v, = 15 ms^{-1} and rotational velocity given by v = $r\omega$. B1

At P, centre velocity and rotational velocity or speed are in the same direction . B1

Resultant velocity = 30 ms^{-1} ~~B1~~

At Q, centre velocity and rotational velocity or speed are directly opposite. Resultant velocity = 0. B1

(d) (i)



By resolution

$T_{bc}\cos 60 = T_{ba}\cos 20$ C1

$T_{bc} = 1.88\ T_{ba}$ ~~C1~~

$2.0 \times 9.81 = T_{bc}\ \text{Sin}\ 60 - T_{ba}\ \text{Sin}\ 20$ C1

$= 1.29T_{ba}$

$\therefore T_{ba} = \dfrac{2.0 \times 9.81}{1.29} = 15.3\ \text{N}$ Accept 15.2 or 2 s.f. A1

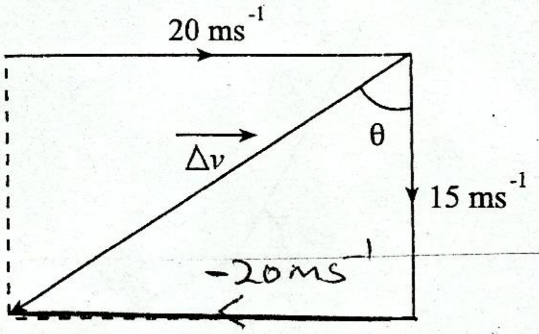
(ii) $T_{bc} = 1.88 \times 15.3$ Accept 28.6 N or 29 N to 2 s.f.

$= 28.7\ \text{N}$ A1

2. (a) Vector has a direction AW B1

Scalar has no direction AW B1

(b) (i)



1. Correct directions of the $20\ \text{ms}^{-1}$ & 15ms^{-1} on diagram B1

2. ΔV on diagram pointing south-Westal direction. B1

(ii)

1. change in speed $= 15 - 20$

$= (-)5\text{ms}^{-1}$ A1

Ignore minus sign

2. change in velocity $\overline{\Delta V} = \sqrt{20^2 + 15^2}$ C1

$= 25\ \text{ms}^{-1}$ A1

at angle $\theta = \tan^{-1}\left(\frac{20}{15}\right)$ or $\tan^{-1}\left(\frac{15}{20}\right)$

$= 53.1°$ (to the vertical) or $36.9°$ to the horizontal A1

(c) Acceleration is change in velocity per unit time. B1

∴ Acceleration has the same direction as the change in velocity/AW Accept examples with explanation. B1

3 (a) (i) a pair of forces equal in size but opposite in direction B1
which act on a single body to cause rotation B1

(ii) Resultant force is zero / AW B1

Resultant torque is zero / AW B1

(b) (i) $8\ T \sin 53° = (600 \times 2) + (200 \times 4)$ C1

$$T = \frac{2\,000}{8 \sin 53°}$$

$= 313$ N A1

(ii) Horizontal component R_h $=$ $313 \cos 53°$ C1

$=$ 188 N ~~C1~~

Vertical component R_v $=$ $600 + 200 - 313 \sin 53°$ C1

$=$ 550 N ~~A1~~

Reaction force R $= \sqrt{550^2 + 188^2}$

$=$ 581 N A1

(iii) $\theta = \tan^{-1}\left(\frac{550}{188}\right)$

$= 71.1°$ A1

(c) Use of a beam balance Accept definition in symbols defined A1

4 (a) Force per unit mass B1

(b) (i) $\frac{mv^2}{r} = \frac{GM_E m}{r^2}$

$\frac{v^2}{r} = \frac{GM_E}{r^2}$ ~~C1~~ B1

$v^2 = \frac{GM_E}{r}$

Also $v = \frac{2\pi r}{T}$ / $\omega = \frac{2\pi}{T}$ ~~C1~~ B1

$\therefore T = 2\pi\sqrt{\frac{r^3}{GM_E}}$ A1

(ii) 1. $g_1 = \frac{R_E^2}{(R_E+h)^2} \times g$

Follow thru candidate's working

$= \left(\frac{6\,400}{8\,400}\right)^2 \times 9.81$ C1.

$= 5.69\ \text{N kg}^{-1}$ Accept 5.67 N kg^{-1} Reject answer in ms^{-2} ~~with units of m~~

2. $v^2 = g_1 r$

mind ecf

$= 5.69 \times 8\,400 \times 10^3$ C1

$v = 6.92 \times 10^3\ \text{ms}^{-1}$ A1

(c) **Satellite needs fuel to keep it in orbit and has to be replaced when fuel runs out/AW/other plausible reason.** B1 B1

Accept :- Satellite lifespan is shortened by friction, loss of energy, technical, ~~faults~~ mechanical & physcal faults, effects of the sun (eclipse) B1

- The Satellite has to be replaced when these effects occur B1

Satellites

The life span of satellites depends largely on their size, or to be precise, it depends on how much liquid fuel they carry aboard. The liquid fuel is used to operate small rocket engines on the satellites. The rocket engines are very important for the satellite because there are three types of forces acting on the satellite in space causing the satellite to deviate from its course.

- before the satellite runs out of fuel , the ground station performs one last manoeuvre to send the satellite to a place called (the satellite graveyard) , it's an orbit around the earth used to retire old useless satellites & they become junk on space , a growing problem , this is when in it come to satellites using the geostationary orbit , for satellites using lower orbits they may be put on re-entry course so they burn before they reach the earth.

- Satellites are big money and made to last but it all depends on the type and use. A satellite's life can end by having it's orbit decayed (or moved by small on board rockets into the atmosphere) and then burned up or it can be simply left in place (becoming space junk a growing problem).

- Life span depends on the spacecraft, besides fuel also on technical failure of electrical components assembled in the satellites and equipment in Earth base stations. If the satellite runs out of fuel and when mechanical and electrical failures occur then it can not station keep and is basically debris.

Another likely scenario that will result in super-syncing a satellite would be loss of mission capability. Again, depending on the severity of the anomaly, the satellite would either continue on with mission, be placed in storage at super-sync or be decommissioned after being super-synced. There is a slight chance that with the loss of mission capability, the customer will choose to keep the satellite in storage at the super-sync orbit to perform longevity studies on the satellite or further testing that can provide insight to other satellites in the constellation or for future builds. The most likely scenario would be to decommission the satellite since the purpose of any satellite is to perform the mission it was designed for. Loss of mission capability includes loss of the payload or payloads, loss of power to the payload, loss of the ability to transmit mission data, loss of the ability to maintain mission attitude.

Once in super-sync and in storage mode, the extended life of the satellite can last for years, even lasting as long as the original design life. This is due to minimal use of fuel and most equipment in an off-state or a safe-state. At this stage of a geo-satellite, the decision to decommission is based on the ability to maintain the satellite in storage and its capability to perform mission when needed. Fuel maintenance has become less significant, since the satellite has already been taken out of the mission orbit. The ability to maintain power levels and attitude become more significant. A geo-satellite will have to be able to maintain power levels while in sunlight and through eclipse. A geo-satellite will experience 2 seasons of eclipse a year that can last anywhere from 30 minutes to 1 hour daily.
Because of this non-daily eclipse throughout the year, the battery will deteriorate more quickly than a low earth orbiting satellite and require more maintenance as the satellite ages. Either the loss of the batteries ability to maintain charge or the excess maintenance of the batteries is both reasons to decommission.
Since the attitude control systems is so complex, various anomalies can result in difficulties in maintaining attitude or maintaining momentum. So losing either capability will affect performing mission and thus can result in decommissioning a satellite.

Minor factors include the ability to maintain communication and maintain operable temperatures. For communication, most satellite design will have enough communication back up in both command plink and telemetry downlink. So communication is usually not a major factor, but if enough anomalies compile on top of each other, then the difficulty of communicating to the satellite might be cause to decommission a satellite. And losing the ability to transmit mission data is obviously a life limiting factor, as mentioned earlier. As for temperatures, as the satellite ages, thermal properties diminish, resulting in higher than ideal temperatures. Higher temperature can affect mission by changing the